

Institute of Actuaries of India

Subject CS1-Actuarial Statistics (Paper A)

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i) $P(\mu > 60) = P(z > (60-65)/17) = P(z > -0.2941)$ [1]
 $= 61.57\%$ [1]

ii) From tables page 28, we know μ/X follows Normal (μ^* , σ^{*2})

$$\mu^* = (150 \cdot 63 / 20^2 + 65 / 17^2) / (150 / 20^2 + 1 / 17^2)$$

$$= 63.02$$
 [1]

$$\sigma^{*2} = 1 / (150 / 400 + 1 / 289) = 2.6423$$
 [1]

So $P(\mu > 60) = P(z > (60-63.02) / \sqrt{2.6423}) = P(z > -1.86)$ [1]
 $= 96.86\%$ [1]

iii) The probability has increased as there is higher certainty over the value of μ as a result of considering a fairly large sample.

This is despite the fact that the mean belief about μ has fallen, which apriori might make a lower value of μ more likely

The posterior distribution has thinner tails and hence lower volatility since the credibility increases around the mean. [2]

[8 Marks]

Solution 2:

i) Let 'u' be the simulated value from the Uniform Distribution. Solving for 'x' in terms of 'u':

$$u = 1 - e^{-(x)^2}$$

$$-(x)^2 = \ln(1 - u)$$

$$x = \sqrt{-\ln(1 - u)}$$

Hence correct answer is 'b' [2]

ii) for $u = 0.75$, $x = \sqrt{-\ln(1 - 0.75)} = 1.177$ [1]

iii) Deriving the probability distribution function:

$$F(x) = \int_0^x (1+t)^{-2} dt = [-(1+t)^{-1}]_0^x$$

$$F(x) = 1 - (1+x)^{-1}$$

Correct answer is 'a' [2]

iv) Equating and Solving for 'u'

$$u = 1 - (1+x)^{-1}$$

$$x = (1 - u)^{-1} - 1$$
 [1]

For $u=0.5$, $x = (1 - 0.5)^{-1} - 1 = 1$

[1]

[7 Marks]

Solution 3:

- i) Spearman rank correlation coefficient r_s is given by $r_s = 1 - \frac{6 \sum d_i^2}{n(n^2-1)}$ correct answer is 'd' [1]
- ii) $d_i = \text{rank of } (X_i) - \text{rank of } (Y_i)$ [0.5]
Condition necessary: there should not be any 'ties' in the rank of variables [0.5]
- iii) A pair of observation $(X_i, Y_i); (X_j, Y_j)$ where $i \neq j$ is said to concordant if $((X_i > X_j) \text{ and } (Y_i > Y_j))$ or $((X_i < X_j) \text{ and } (Y_i < Y_j))$ [1]
- iv) Kendall rank correlation coefficient τ is given by $\tau = \frac{n_c - n_d}{n(n-1)/2}$ correct answer is 'b' [1]
- v) Where n_c is the number of concordant pair in the data, [0.5]
 n_d is the number of discordant pair in the data and [0.5]
 n is the number of observations [0.5]
Condition necessary: there should not be any 'ties' in the rank of variables [0.5]

[6 Marks]

Solution 4:

- i) Option a [3]
- ii) $f(p/x)$ is proportional to $f(x/p) * f(p)$ [0.5]
 $= p^d * (1-p)^{(n-d)} * p^{(\alpha-1)} * (1-p)^{(\beta-1)}$
 $= p^{(\alpha+d-1)} * (1-p)^{(n-d+\beta-1)}$ [1.5]
Which corresponds to the pdf of Beta distribution with parameters $(\alpha + d)$ and $(\beta + n - d)$ [1]
- iii) Option d [3]
- iv) Option a [3]
- v) Higher σ^2 means high volatility and less certainty in the prior estimate. Prior becomes less reliable and leads to higher weight to observed data and hence higher credibility factor. [1]

[13 Marks]

Solution 5:

- i) Since only certain part of experience is given, we should use truncated Poisson distribution: *(one can use L to mean λ)*
 $P(X = x) = k \frac{L^x \exp(-L)}{x!}$ [1]
where $x = 2, 3, 4, \dots$
and k is constant such that the sum of probabilities = 1 [0.5]
We also have that $P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$
 $P(X \geq 2) = 1 - \exp(-L) - L \cdot \exp(-L)$ [1]
Since $k \sum_{x=2}^{\infty} P(X = x) = 1 \Rightarrow k(1 - \exp(-L) - L \cdot \exp(-L)) = 1$ [1]
Or $k = 1/(1 - \exp(-L) - L \cdot \exp(-L))$ [0.5]

Hence the function is given by:

$$P(X = x) = \frac{L^x \exp(-L)}{x!} \cdot \frac{1}{(1 - \exp(-L) - L \exp(-L))}$$

ii) The maximum likelihood function is given by:

$$L(\lambda) = \prod_{i=2}^n \frac{\lambda^{x_i} e^{-\lambda}}{x_i! (1 - e^{-\lambda} - \lambda e^{-\lambda})} = \text{constant} \times \frac{\lambda^{\sum x_i} e^{-\lambda(n-1)}}{(1 - e^{-\lambda} - \lambda e^{-\lambda})^{(n-1)}} \quad [1]$$

$$\Rightarrow \log(L(\lambda)) = \text{constant} + \sum x_i \log(\lambda) - \lambda(n-1) - (n-1) \log(1 - e^{-\lambda} - \lambda e^{-\lambda}) \quad [1]$$

Differentiating wrt λ

$$\frac{d}{d\lambda} \log(L(\lambda)) = \frac{\sum x_i}{\lambda} - (n-1) - \frac{(n-1)(e^{-\lambda} + \lambda e^{-\lambda} - e^{-\lambda})}{(1 - e^{-\lambda} - \lambda e^{-\lambda})} \quad [1]$$

$$\Rightarrow \frac{d}{d\lambda} \log(L(\lambda)) = \frac{n\bar{x}}{\lambda} - (n-1)(1 - e^{-\lambda}) / (1 - e^{-\lambda} - \lambda e^{-\lambda}) \quad [1]$$

Equating to zero,

$$0 = \frac{n\bar{x}}{\lambda} - (n-1)(1 - e^{-\lambda}) / (1 - e^{-\lambda} - \lambda e^{-\lambda}) \quad [0.5]$$

$$\Rightarrow \frac{n\bar{x}}{\lambda} = \frac{(n-1)(1 - e^{-\lambda})}{(1 - e^{-\lambda} - \lambda e^{-\lambda})} \Rightarrow \bar{x} = \frac{\lambda(n-1)(1 - e^{-\lambda})}{n(1 - e^{-\lambda} - \lambda e^{-\lambda})} \quad [0.5]$$

iii) From the given data, $\bar{x} = \frac{230 \times 2 + 54 \times 3 + 6 \times 4}{230 + 54 + 6} = 2.2275$ [1]

With λ as 0.6, RHS of result in part (ii) is 2.2131 [1]

With λ as 0.7, RHS of result in part (ii) is 2.2539 [1]

Using linear interpolation,

$$\lambda = 0.6 + \frac{2.2275 - 2.2131}{2.2539 - 2.2131} (0.7 - 0.6) = 0.635 \quad [1]$$

[13 Marks]

Solution 6:

We need to perform χ^2 contingency test with:

H_0 : There is no association between travel history and symptoms vs

H_1 : There is some association between travel history and symptoms [1]

Total number of observations are 500. [1]

The expected frequencies are:

	No Symptoms (NS)	Some Symptoms (SS)	Confirmed case (CC)
Travelled to place outside Actuarial (T)	82.51	66.09	42.40
No travel in recent past (NT)	133.49	106.91	68.60

[3]

The test statistic is

$$\sum \frac{(O-E)^2}{E} = \frac{(86-82.51)^2}{82.51} + \dots + \frac{(31-68.80)^2}{68.80} = 95.74 \quad [2]$$

This is much higher than the critical value of 9.21 at 1% confidence. Hence there is enough evidence to conclude that there is some association between the travel history and the virus symptoms. [1]

[8 Marks]

Solution 7:

i) Correct answer is 'c' [1]

ii) Correct answer is 'b' [1]

iii) $E(e^{tX}) = \frac{1}{\theta} \left(\frac{1}{\theta} - t \right)^{-1} \int_0^{\infty} \exp(-z) \cdot dz$
 $E(e^{tX}) = (1 - t\theta)^{-1}$
 Correct answer is 'd' [2]

iv) The total time y_i to count N buses will be $y_i = x_1 + x_2 \dots + x_N$ [1]

MGF of Y is given by $E(e^{tY}) = E(e^{t \sum X}) = \prod_1^N e^{tX}$ [1]

$M_Y(t) = (1 - t\theta)^{-N}$ [1]

v) The $M_Y(t)$ is of the form of MGF for *Gamma* (α, λ) .

Hence the distribution in our case is *Gamma* $\left(N, \frac{1}{\theta}\right)$

[2]

[9 Marks]

Solution 8:

i) Null hypothesis: There is no difference in variance of number of visitors in cold week and number of visitors in a warm week

Test statistic: $S_{\text{cold}}^2 / S_{\text{warm}}^2 \sim F_{24,34}$ [1]

$S_{\text{cold}}^2 = 1/24 * (4,718,190 - 25 * (10,250/25)^2) = 21,487.08$ [1]

$S_{\text{warm}}^2 = 1/34 * (4,740,201 - 35 * (12,001/35)^2) = 18,389.10$ [1]

The 95% quantile of $F_{24,34}$ is 1.843 and observed value is

$F = 21,487.08 / 18,389.10 = 1.17 < 1.83$ [1]

Hence there is no evidence that variance of visitors is higher in colder weeks [1]

ii) Null hypothesis: There is no difference in mean number of visitors in cold and warm week

Assuming that the two population variances are equal, we have:

$S_p^2 = (24 * 21,487.08 + 34 * 18,389.10) / 58 = 19,671.02$ [2]

$t = (10,250/25 - 12,001/35) / \sqrt{19,671.02 * (1/35 + 1/25)} = 1.8274$ [1]

The 97.5% (two sided test) of the t_{58} is 2.009 and 2.000.

Hence there is no evidence suggesting that the mean number of visitors differ between cold and warm weeks. [1]

Alternatively, students can use the z statistic considering that samples are large:

$$z = (10,250/25 - 12,001/35) / \sqrt{(21,487.08/25 + 18,389.10/35)} = 1.8035$$

Since this is lower than 1.96, null hypothesis is accepted and same conclusion as t test.

iii)

Option 1 (considering number of weeks as 78):

$$S_{tx} = 582,205 - 1/78 * 1,603 * (10,250 + 12,001) = 124,918.42 \quad [1]$$

$$S_{xx} = (4,718,190 + 4,740,201) - (10,250 + 12,001)^2 / 78 = 3,110,865.35 \quad [1]$$

$$S_{tt} = 51,823 - 1,603^2 / 78 = 18,879.29 \quad [1]$$

$$\text{Rho}(T,X) = 124,918.42 / \sqrt{(3,110,865.35 * 18,879.29)} = 0.5155 \quad [1]$$

This shows a weak positive correlation between the number of visitors and temperature, i.e. number of visitors increase with increase in temperature, however the correlation is weak [1]

Option 2 (considering number of weeks as 60):

$$S_{tx} = 582,205 - 1/60 * 1,603 * (10,250 + 12,001) = -12,267.55 \quad [1]$$

$$S_{xx} = (4,718,190 + 4,740,201) - (10,250 + 12,001)^2 / 60 = 1,206,607.65 \quad [1]$$

$$S_{tt} = 51,823 - 1,603^2 / 60 = 8,996.18 \quad [1]$$

$$\text{Rho}(T,X) = -12,267.55 / \sqrt{(1,206,607.65 * 8,996.18)} = -0.12 \quad [1]$$

This shows a small negative correlation between the number of visitors and temperature, i.e. number of visitors increase with drop in temperature, however the correlation is quite small to come to any concrete conclusions. [1]

iv) **Option 1 (considering number of weeks as 78):**

$$A = (78 * 582,205 - 1,603 * (10,250 + 12,001)) / (78 * 51,823 - 1,603^2) = 6.6167$$

Or from part iii),

$$A = 124,918.42 / 18,879.29 = 6.6167$$

$$B = ((10,250 + 12,001) - 6.6167 * 1,603) / 78 = 149.29 \quad [2]$$

Option 2 (considering number of weeks as 60):

$$A = (60 * 582,205 - 1,603 * (10,250 + 12,001)) / (60 * 51,823 - 1,603^2) = -1.3636$$

Or from part iii),

$$A = -12,267.55 / 8,996.18 = -1.3636$$

$$B = ((10,250 + 12,001) - (-1.3636) * 1,603) / 60 = 407.29 \quad [2]$$

[16 Marks]

Solution 9:

- i) Assumption : the sample is from a normal population. [0.5]

For a sample from normal distribution, we know that $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

$$\bar{X} \sim N\left(2000, \frac{500^2}{10}\right) \quad [0.5]$$

$$P(\bar{X} < 1700) = P\left(Z < \frac{1700-2000}{\frac{500}{\sqrt{10}}}\right) \quad [1]$$

$$= P(Z < -1.8974) = 1 - \Phi(1.8974) = 1 - 0.9711 = 2.889 \% \quad [1]$$

- ii) Assumption : the sample is from a normal population. [0.5]

For a sample from normal distribution, we know that $\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$ [0.5]

$$P(S^2 < 250^2) = P\left(\frac{(10-1)S^2}{500^2} < (10-1) \frac{250^2}{500^2}\right) = P(\chi_9^2 < 2.25) = 1.313 \% \quad [2]$$

- iii) Assumption : the sample is from a normal population and hence the mean and variance are independent. [0.5]

$$P(\bar{X} < 1700 \cap S^2 < 250^2) = P(\bar{X} < 1700) \cdot P(S^2 < 250^2) = 0.02889 \times 0.01313 = 0.038\% \quad [0.5]$$

[7 Marks]

Solution 10:

- i) A saturated model has as many parameters as there are data points and would therefore be a perfect fit to the data. It is not useful to predict and hence not used in practice. [2]

ii)

- a) Pearson residuals are $(y_i - \mu_i) / \sqrt{\text{var}(\mu_i)}$ where μ_i is the fitted response estimator. [1]

The deviance residuals are $\text{sign}(y_i - \mu_i) d_i$ where d_i is the contribution of the i-th to total deviances, sum of d_i^2 is the scaled deviance. [1]

- b) The Pearson residuals tend to be skewed in non normal data while deviance residuals tend to be symmetric and hence normal assumption is more appropriate. For that reason, the latter is preferred in actuarial applications [2]

- c) For normally distributed data, the Pearson and deviance residuals are identical [1]

- iii) Using the information given, we can calculate the deviance differences and compare that with the differences of the degrees of freedom for each of the model. If the decrease in the deviances is greater than twice the difference in degrees of freedom this suggests an improvement. [1]

Model	Choice of predictor	Degree of freedom	Scaled Deviance	Difference of scaled deviance
A	1	16	900	
B	Age	13	800	100
C	Age + BMI	10	770	30
D	Age * BMI	1	760	10

[4]

From the above table, it is seen that interaction model does not indicate significant improvement and hence the recommended model is Age + BMI. [1]

[13 Marks]
